Performance of sPEEK Catalyst in low-temperature Solketal-Synthesis

Alina Greve [a,b], Hendrik Stein[b], Thomas Osterland[c], Olaf Hinrichsen[a,d]

[a]TUM School of Natural Sciences, Department of Chemistry, Technical University of Munich, Garching/Germany; [b]ASG Analytik-Service AG, Neusäss/Germany; [c]Faculty of Mechanical and Process Engineering, Technical University of Applied Sciences Augsburg, Augsburg/Germany; [d] Technical University of Munich, Catalysis Research Center, Garching/Germany

Scope and motivation

In previous studies, sulfonated poly (ether ether) ketone (sPEEK) showed good activity in esterifications and etherifications as a non-halogenated solid-acid catalyst in the temperature range of 110°C - 170 °C. [1] For these reactions, the kinetics would be too slow to decrease the temperature below 100°C.

This can be different for other types of reactions, such as acetylations, as has been shown for different commercially available catalysts. [2] To investigate the performance at low temperatures, sPEEK was synthesized and tested in the acetylation of glycerol and acetone to solketal ($C_6H_{12}O_3$) at temperatures between 30 °C and 50 °C. The scope of this work is to explore further applications for sPEEK as a solid-acid catalyst and tailor the material to obtain a chemically stable, and highly active material for acid-catalyzed reactions.

Materials and methods

For sulfonation with concentrated sulfuric acid, the polymeric backbone VICTREX™ PEEK 450G™ (Victrex plc) was used to synthesize sPEEK as a catalyst. [3] Reaction time varies between 30 to 360 minutes, and reaction temperature is between 40 °C and 60 °C. The catalyst was neutralized and washed with deionized water. After drying, the resulting ion-exchange capacity (IEC) and degree of sulfonation (DS) of the catalyst were analyzed by back titration with hydrochloric acid.

Acetylation of glycerol and acetone was carried out in a batch reactor (V = 250 mL) to test the catalytic activity and selectivity. Glycerol and acetone were used in ratios between 1:5 and 1:20 (mol:mol) with a catalyst-to-substrate ratio of 1.2 wt-%, respectively. Considering the boiling point of acetone at 56 °C, the experiments were

performed at reaction temperatures of 30 °C and 50 °C at atmospheric pressure. The reaction product was analyzed by gas chromatography (GC-FID).

Results and discussion

Variations of sPEEK catalyst with an IEC between 0.3 and 2.5 mmol/g, corresponding to DS between 10 % and 90 %, were synthesized. First results of glycerol conversion after acetylation reaction are shown in Figure 1. H-sPEEK-87 (IEC: 2.3 mmol/g) has the highest performance of the tested catalysts. Although Lewatit® GF101 has a higher IEC (4.7 mmol/g) than H-sPEEK-11 (IEC: 0.4 mmol/g), H-sPEEK-11 shows better performance. The activity of Amberlyst™ 36 is comparable to that of sPEEK with a sulfonation degree higher than 21 %.

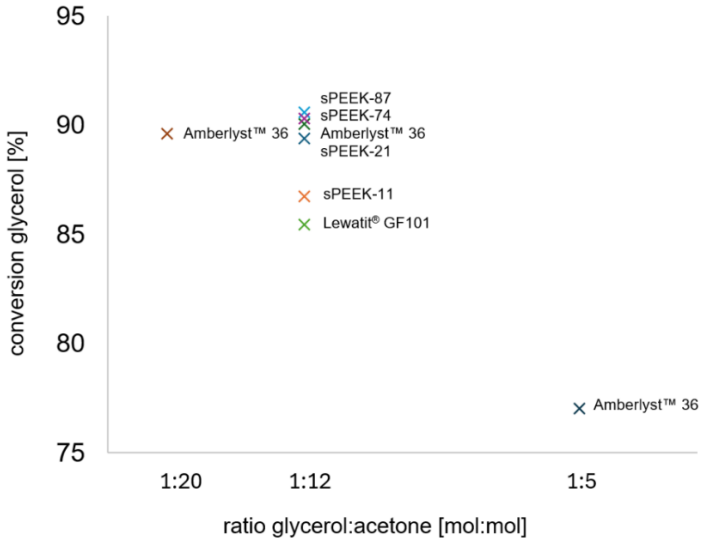

Figure 1: Comparison of the commercial catalysts Lewatit® GF101 and Amberlyst™ 36 with sPEEK catalysts for the conversion of glycerol to solketal at 40 °C for 60 min and 1.2 wt-% catalyst, respectively. Conversion of glycerol without catalyst (t = 540 min, T = 40 °C, ratio 1:12): 42 %;

Conclusion

Even at low temperatures, sPEEK showed a high activity in the synthesis of solketal with glycerol and acetone as a non-halogenated solid-acid catalyst. Due to modifications of the catalysts' macrostructure and adjustments in catalyst synthesis, further improvements for catalyst activity and selectivity are intended.

References

[1] A. Greve, H. Stein, T. Osterland, O. Hinrichsen, *Applied Catalysis O: Open* **2024**, *190*, 206951. DOI: https://doi.org/10.1016/j.apcato.2024.206951

[2] J. Esteban, F. García-Ochoa, M. Ladero, *Green Processing and Synthesis* **2017**, *6 (1)*, 79 – 89. DOI: https://doi.org/10.1515/gps-2016-0105

[3] N. Shibuya, R. S. Porter, *Macromolecules* **1992**, *25 (24)*, 6495 – 6499. DOI: https://doi.org/10.1021/ma00050a017